

ESC provides varying electric power from the battery pack, according to the input signal. Most ESCs also have a battery eliminated circuit, which is nothing but a 5V DC output that can power-up the electronics, including the receiver, servo motor and flight controller.

Another feature available in many ESCs is low-voltage cut-off (LVC) circuitry. This protects the LiPo battery pack from being discharged too much, to save it from being permanently damaged. We can fine-tune most ESCs with exotic throttle profiles (throttle range, battery mode, etc), with the help of computer based configuration programs.

Rough calculation. If the maximum current rating of the motor is 10A, then the capacity of the ESC should be $10 \times 1.5 = 15A$ and $I_{(ESC)} = I_{max} (motor) \times 1.5$

The propeller size strongly influences the electric current flow through a system. A large propeller will draw higher current from the battery through the ESC and make the system run hot. Since every motor, ESC and battery operates at a specific maximum allowable current, exceeding the limits could soon burn-out the motor, ESC or battery pack.

Battery pack

Compared to NiMH and NiCd battery packs, LiPo battery packs are ideal for use with brush-less motors in radio-controlled airplanes and copters due to their small profile, low weight and high capacity. When looking for a battery pack, keep an eye on the maximum current rating of the motor(s) used, at first.

Some important specifications of a battery pack are mAh, C and 1S/2S/3S/4S.

mAh rating indicates the capacity of the battery pack (more mAh gives more flight time). C is the discharge rate, that is, an indicator of the maximum current (A) level at which a battery can be discharged at a particular time period. 1S/2S/3S/4S reveals the inside combination of the LiPo battery



Electronic speed controls



LiPo battery pack



Flight controller

pack. 1S means one cell of 3.7V, while 4S means four cells of 3.7V (14.8V). Just multiply C rating with the battery pack's capacity (mAh) to determine how many amperes can safely be pulled from the battery pack. For example, an 800mAh battery with 10C rating can continuously provide an output of $8A$ ($0.8 \times 10 = 8$).

Quick tip. For a longer flight time, it is better to use a battery with higher mAh capacity at the same C rating, or a battery with higher C rating at the same mAh capacity. For example, an 800mAh 20C battery is better than 1500mAh 10C battery, because $800mAh \times 20C = 16A$ and $1500mAh \times 10C = 15A$.

Flight controller (FC)

Literally, a flight controller overlooks the operation of the flying machine,

including reading RC radio inputs, controlling motors and determining the orientation.

RC radios have multiple signal outputs, one for each channel/stick/switch/knob. Each radio output transmits a pulse (usually at 50Hz), with the width of the pulse determining where the stick is on the RC transmitter (for example, 1000µs for throttle 0).

As stated, motors are controlled with the ESC. It is always necessary to determine which orientation, or attitude, the flying machine is in for, for varying the motor speed along with the pilot's commands. In practice, two sensors are commonly used for determining orientation—accelerometer and gyroscope. An accelerometer measures acceleration in each direction and a gyroscope measures the angular velocity.

DIYers, think first

Building your own ESC for brushed/brush-less motors is not a very difficult task if you have enough knowledge, skill and patience. There are a number of tried and tested designs that have been published for ESCs. However, reliability of a home-made ESC may raise an important safety issue because a DIY ESC is probably going to be heavier than an SMT based modern commercial ESC. Under certain circumstances, you can save money by building your own ESC, but most of the time there is not much cost advantage to it.

Important challenges in designing an ESC are:

1. Accurate timing of the throttle control pulse
2. Accurate pulse width modulation (PWM) generation to ensure the motor speed is even, and the motor does not make unacceptable noise and harmful vibrations
3. Integration of optional features, like lost model alarm (LMA) and glitch counter
4. Maximising power efficiency

Warning! Model aircraft is not a toy. Please construct, test and use RC flying models at your own risk. ●